Idea 1: Audio

Idea 2: Healthcare time series

Standard models (like LSTMs) predict the average future. In healthcare, the average is useless. If a patient has a 50% chance of stabilizing and a 50% chance of cardiac arrest, an LSTM predicts a "stable but slightly weird" middle ground that exists in neither reality.

Diffusion models, however, can generate 50 different possible futures for a single patient. If 10 of those futures show a crash, you trigger an alarm.

The Project: "The Probabilistic Digital Twin"

The Goal: Build a model that takes the last 24 hours of a patient's vitals (Heart Rate, Blood Pressure, SpO2) and "hallucinates" the next 2 hours of potential outcomes to detect early deterioration risks.

1. The Architecture

You will likely use CSDI (Conditional Score-based Diffusion Imputation) or TimeGrad. These models treat "forecasting" as an "imputation" problem where the "future" is just missing data to be filled in.

Input: A tensor of shape (Batch, Vitals, Time_Steps) (e.g., (32, 3, 100)).

Conditioning: The known past (t=0 to t=80).

Diffusion Task: The model receives pure noise for the future (t=80 to t=100) and iteratively denoises it, conditioned on the past, to reveal a plausible future.

2. The Workflow

Data Ingestion: Download the MIMIC-III dataset (open-access ICU data). Extract timeseries for Heart Rate (HR), Mean Arterial Pressure (MAP), and Respiratory Rate (RR).

Preprocessing: Normalize the data. Create "sliding windows" of 12 hours. Mask the last 2 hours of every window to serve as your "ground truth" target.

Training: Train a CSDI model to reconstruction the masked 2-hour window.

Inference (The "Twin" Part): For a new patient, run the reverse diffusion process 50 times.

Risk Analysis:

Scenario A: All 50 generated futures stay stable. -> Low Risk.

Scenario B: 15 out of 50 futures show a drop in Blood Pressure < 60 mmHg. -> High Risk Alarm.

3. Why this is better than standard AI

A standard regression model outputs a single line. If the patient is volatile, that line will be flat (regression to the mean). A diffusion model preserves the variance, showing you the "worst-case scenario" explicitly.
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Fine, Nan. I'll give you a **clean, professor-friendly project proposal**, but I'm keeping the mild attitude because otherwise you'll think I suddenly became soft.

Below is your polished, structured, research-scientist-level proposal for **"The Probabilistic Digital Twin"**—combining the two ideas you liked (diffusion time-series forecasting + AE-diffusion anomaly detection) without turning it into a 3-PhD-theses monstrosity.

Use this tomorrow. Your prof will understand it instantly.

Project Proposal: The Probabilistic Digital Twin

One-Line Summary

A conditional diffusion model generates multiple plausible future vital-sign trajectories for ICU patients, enabling early detection of deterioration by analyzing risk across *distributions* instead of point predictions.

Where It Comes From (Real Research Basis)

- **Time-series diffusion forecasting literature** (e.g., CSDI, TimeGrad) shows diffusion models excel at *probabilistic* multi-step forecasting—future = masked/imputed region.
- **AE + diffusion anomaly detection** research demonstrates that diffusion models reconstruct the "manifold of normal," exposing deviations as early-warning signals.
- Medical time-series papers repeatedly mention the need for **uncertainty-aware forecasting**, because deterministic models (LSTM, GRU, TCN) collapse volatile patient futures into meaningless averages.

This project sits at that intersection:
Forecast variability + anomaly scoring = actionable early deterioration warning.

Core Approach (Technical + Plain English)

Input

Last 12–24 hours of vitals:

- Heart Rate (HR)
- Mean Arterial Pressure (MAP)
- SpO2 or Respiratory Rate (RR)

Tensor shape: **(Batch, Features=3, Time)**, e.g., (32, 3, 240).

Architecture

1. Encoder (Optional but Lightweight)

A small **1D Convolutional Autoencoder** maps vitals into a compact latent space.
Why?

- Cleans noise.
- Makes diffusion faster (latent dimension << time dimension).
- Follows the "AE + diffusion" hybrid line of research.

2. Conditional Diffusion Model (CSDI-style)

- Treat the **future** as **missing data to impute**.

- Condition on the past 10–22 hours.
- Inject pure Gaussian noise into the masked future window.
- Run reverse diffusion steps to generate plausible future trajectories.

3. Multi-Sample Inference (Digital Twin Simulation)

- Run the diffusion model **N = 30–100** times for one patient.
- Each run gives a different but statistically valid future.

4. Risk Scoring Module

Simple statistical detector:

- Probability of MAP < 60
- Probability of SpO2 < 90
- Probability of HR > 130

If **≥ threshold** of runs show dangerous profiles, raise an alert.

Plain English:
Instead of predicting *one* future, we spawn 50 parallel universes of the patient. If too many universes look bad, we treat the patient as high-risk.

Innovation / Novelty

1. **Distribution-aware forecasting instead of deterministic prediction**
    - Standard models output a single averaged future.
    - Diffusion outputs a full distribution of possible futures.
2. **Hybrid AE + diffusion architecture**
    - Improves robustness.
    - Matches future-work directions in anomaly detection research.
3. **Clinical-style risk scoring**
    - Outputs interpretable risk probabilities (“12/50 futures show MAP collapse”).
    - More actionable than a single regression curve.

4. **Lightweight + implementable**
    - Avoids full transformer-based forecasting beasts.

Feasibility (1 Month, 3 Students, UB MS ES-AI)

- **Training footprint:** Small 1D models (easy on Colab).
- **Dataset:** Publicly accessible MIMIC-III or PhysioNet ICU waveforms.
- **Tasks are modular:**
    - Person 1: preprocessing + feature extraction
    - Person 2: AE + diffusion model training
    - Person 3: inference sampling + risk analysis/visualization
- You do **not** need huge GPU clusters or 3D medical imaging pipelines.

You can realistically finish an MVP and present strong results.

MVP Scope (What You Actually Build)

1. Dataset Pipeline

- Download a subset of MIMIC-III Waveform or PhysioNet ICU data.
- Extract 3 vitals: HR, MAP, SpO2/RR.
- Resample to 5-minute intervals.
- Normalize per-vital.
- Create sliding windows:
    - Input: first 10 hours
    - Target future: next 2 hours

2. Model Training

- Train an autoencoder to compress each 3×120 window into latent z.
- Train a **CSDI-style conditional diffusion model** on z to impute the last 2 hours.

3. Digital Twin Simulation

- For each patient window, generate 50 future trajectories.
- Plot distribution envelopes (fan charts).

4. Risk Detector

- Basic rules like:
    - MAP < 60 for ≥ X% of generated futures
    - SpO2 < 92 for ≥ X% of futures
- Output binary risk label + probability of deterioration.

5. Evaluation

- Compare against baselines:
    - LSTM/GRU deterministic forecast
    - Simple ARIMA (for fun)
- Metrics: CRPS, MAE, spread vs. sharpness, anomaly recall.

Everything above is doable with 1–2 GPUs on Colab.

Why This Beats Standard AI (Professor-Friendly Section)

Deterministic model problem

LSTMs predict **the mean** of unstable trajectories.
Outcomes become “washed out” and clinically meaningless.

Example:

- 50% chance of cardiac deterioration
- 50% chance of stabilization
  → LSTM predicts “slightly abnormal but stable.” Useless.

Diffusion advantage

Diffusion models preserve variance.
They generate **multiple valid futures**, revealing:

- How volatile a patient really is
- Early warning tails (“worst-case paths”)
- Probability of rare-but-deadly events

This is exactly what healthcare needs:
not a prediction, but a risk envelope.

High-Level Workflow (for Slides)

1. **Extract ICU vitals**
2. **Preprocess + sliding windows**
3. **Encode to latent space (AE)**
4. **Train conditional diffusion on latent time-series**
5. **Generate 30–100 future trajectories**
6. **Compute deterioration probability**
7. **Trigger alarm if risk > threshold**

Tech Stack

Core

- PyTorch
- PyTorch Lightning (optional, but nice for training loops)
- Diffusers (for some utilities, though you’ll write custom 1D UNet)

Data & Preprocessing

- PhysioNet MIMIC-III / MIMIC-IV
- Pandas / NumPy
- WFDB library for waveform extraction

Visualization

- Matplotlib / Seaborn
- Plotly for interactive fan charts

Compute

- Google Colab Pro or Pro+
- 12–18 GB GPU is enough

Deliverables You Can Promise

- Working diffusion model for ICU vitals
- Demo notebook generating 50 futures for a patient
- Risk scoring module with visual charts
- Comparison against LSTM baseline
- Project paper + poster for class/demo day